Derivative Securities: Final Exam

Derivative Securities (University of Melbourne)



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Derivative Securities

Derivative Securities: University Final Examination

Pages	Total Marks	Section	Questions
12	100	3	26

Section 1 Multiple Choice	15 questions	**30 total marks**
Section 2 Short Answer	8 questions	**40 total marks**
Section 3 Long Answer	3 questions	**30 total marks**

Name _____ Student ID _____

Date _____ Signature _____

How to use this exam

1. Set a timer. Practice under real exam conditions! No notes, no AI assistance.
2. When finished, scan the QR code on the last page to submit and see the correct answers with detailed explanations.
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Section 1 / 3

30 total marks

Multiple Choice

Select the single best answer for each of the following 15 questions. Each question is worth 2 marks.

Question 1

2 marks

A European call option and a European put option on the same underlying stock have the same strike price of $50 and the same expiration date. The current stock price is $48, and the risk-free rate is 5% per annum. If the put price is $4.50 and the call price is $3.50, which of the following actions constitutes a profitable arbitrage strategy?

- **A** Sell the call, buy the put, buy the stock, and borrow at the risk-free rate.
- **B** Buy the call, sell the put, buy the stock, and borrow at the risk-free rate.
- **C** Buy the call, sell the put, sell the stock, and invest at the risk-free rate.
- **D** Sell the call, buy the put, sell the stock, and invest at the risk-free rate.

Question 2

2 marks

Which of the following best describes the 'marking-to-market' process in futures contracts compared to forward contracts?

- **A** It eliminates basis risk for the hedger by locking in the spot price daily.
- **B** It occurs only at the maturity of the contract to settle the final price.
- **C** It involves the daily settlement of gains and losses, significantly reducing counterparty credit risk.
- **D** It is a requirement for OTC forward contracts to ensure liquidity.

Question 3

2 marks

Under the Black-Scholes-Merton model, what happens to the Delta of an in-the-money European Put option as it approaches maturity, assuming the stock price remains constant?

- **A** It approaches +1.
- **B** It approaches 0.
- **C** It remains constant at -0.5.
- **D** It approaches -1.

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Question 4

2 marks

A 'Contango' market for a non-dividend-paying commodity implies that the futures price is:

- **A** Equal to the expected future spot price.
- **B** Higher than the current spot price.
- **C** Lower than the current spot price.
- **D** Decreasing over time as the delivery date approaches.

Question 5

2 marks

Which Greek measures the sensitivity of an option's Delta to changes in the underlying asset price?

- **A** Rho
- **B** Gamma
- **C** Vega
- **D** Theta

Question 6

2 marks

An American call option on a non-dividend-paying stock should:

- **A** Always be exercised early if it is deep in-the-money.
- **B** Never be exercised early.
- **C** Be exercised early if volatility is very low.
- **D** Be exercised early only if the risk-free rate is zero.

Question 7

2 marks

In a risk-neutral world, the expected return on all investment assets is:

- **A** The risk-free rate plus a volatility premium.
- **B** The market rate of return.
- **C** The risk-free rate.
- **D** Zero.

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Question 8

2 marks

What is the primary difference between an Asian option and a standard European option?

- **A** Asian options have two strike prices.
- **B** Asian options are only traded on Asian exchanges.
- **C** Asian options can be exercised at any time before maturity.
- **D** The payoff of an Asian option depends on the average price of the underlying over a period.

Question 9

2 marks

Using the one-step binomial model, if the stock price S can move up to Su or down to Sd, the risk-neutral probability p is calculated as:

- **A** $p = (e^{rT} - u)/(d - u)$
- **B** $p = (d - e^{rT})/(d - u)$
- **C** $p = (e^{rT} - d)/(u - d)$
- **D** $p = (u - e^{rT})/(u - d)$

Question 10

2 marks

A 'Straddle' strategy involves:

- **A** Buying two calls and one put.
- **B** Selling a call and buying a put with the same strike price.
- **C** Buying a call and a put with the same strike price and expiration.
- **D** Buying a call and a put with different strike prices.

Question 11

2 marks

Itô's Lemma is used in derivative pricing to:

- **A** Find the differential of a function of a stochastic process.
- **B** Solve linear programming problems for hedging.
- **C** Calculate the margin requirements for cleared swaps.
- **D** Determine the optimal exercise boundary for American options.

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Question 12

2 marks

A 'Short Hedge' is most appropriate for:

- **A** A producer who already owns an asset and plans to sell it in the future.
- **B** An arbitrageur looking for price discrepancies between two exchanges.
- **C** An investor who expects the market volatility to decrease.
- **D** A company that needs to buy an asset in the future.

Question 13

2 marks

Implied volatility is the volatility that, when used in the Black-Scholes model, produces:

- **A** A delta of 0.5.
- **B** The historical standard deviation of the stock.
- **C** A theoretical price equal to the intrinsic value.
- **D** The current market price of the option.

Question 14

2 marks

In the context of the 2008 Financial Crisis and subsequent regulations, a 'Central Counterparty' (CCP) aims to:

- **A** Eliminate market risk for all participants.
- **B** Set the interest rates for OTC derivatives.
- **C** Act as the buyer to every seller and the seller to every buyer to reduce systemic risk.
- **D** Provide unlimited liquidity to the banking sector.

Question 15

2 marks

A 'Bull Spread' created using call options is constructed by:

- **A** Selling two calls with different strike prices.
- **B** Selling a call with a low strike price and buying a call with a high strike price.
- **C** Buying a call and a put with the same strike price.
- **D** Buying a call with a low strike price and selling a call with a high strike price.

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Section 2 / 3

40 total marks

Short Answer

Answer all 8 questions. Show all calculations and provide concise explanations. Each question is worth 5 marks.

Question 16

5 total marks

The current spot price of gold is $2,000 per ounce. The risk-free rate is 4% per annum with continuous compounding. The storage costs for gold are $10 per ounce per year, payable at the end of the year.

A | Calculate the 1-year forward price of gold, assuming no arbitrage opportunities exist.

Question 17

5 marks

Suppose a stock price S follows a Geometric Brownian Motion: $dS = \mu S dt + \sigma S dz$. Define $G = \ln(S)$. Use Itô's Lemma to derive the process followed by G.

Question 18

5 total marks

A stock is currently trading at $100. In each of the next two 6-month periods, it is expected to increase by 10% or decrease by 10%. The risk-free rate is 5% per annum with continuous compounding.

A | Calculate the price of a 1-year European put option with a strike price of $105 using a two-step binomial tree.

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Question 19

5 marks

Explain the concept of 'Delta Hedging' for a financial institution that has sold a large number of call options. What is the main risk associated with maintaining a delta-neutral position?

Question 20

5 marks

An airline company wants to hedge against rising fuel prices using futures contracts. The standard deviation of quarterly changes in fuel prices is $0.12 per gallon. The standard deviation of quarterly changes in the futures price is $0.14 per gallon. The correlation between the changes is 0.85. Calculate the minimum variance hedge ratio.

Question 21

5 marks

Contrast the Monte Carlo simulation method with the Finite Difference method for pricing derivatives. In what scenarios is Monte Carlo preferred?

Question 22

5 marks

Explain how a 'Binary Call' option (also known as a Digital Call) differs from a standard European Call option in terms of its payoff and its Delta near maturity.

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Question 23

5 marks

Define 'Value at Risk' (VaR). What are the two main parameters that must be specified when reporting a VaR figure?

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Section 3 / 3

30 total marks

Long Answer

Provide comprehensive answers to the following 3 questions. Show all logical steps and derivations. Each question is worth 10 marks.

Question 24

10 total marks

An investment bank has a portfolio of OTC derivatives on a specific stock. The portfolio currently has a Delta of $-5,000$ and a Gamma of $-10,000$. Two traded options are available:

- Option A: Delta $= 0.6$, Gamma $= 2.0$
- Option B: Delta $= 0.5$, Gamma $= 1.5$

The bank wants to make the portfolio both Delta-neutral and Gamma-neutral.

A | 1. Calculate the number of units of Option A and the underlying stock required to achieve a Delta-Gamma neutral position.

2. Explain why achieving Gamma neutrality is important in addition to Delta neutrality for a portfolio with large derivative exposures.

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Question 25

10 total marks

A stock price is currently \$50. The risk-free rate is 5% and the volatility is 30% per annum. You are analyzing a 6-month European call option with a strike of \$52. Using the Black-Scholes formula:

$$d_1 = \frac{\ln(S_0/K)+(r+\sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

$$C = S_0 N(d_1) - Ke^{-rT}N(d_2)$$

A

1. Calculate d_1 and d_2 for this option.

2. If $N(d_1) = 0.46$ and $N(d_2) = 0.38$, calculate the price of the call.

3. Discuss how an increase in volatility from 30% to 40% would affect the price of this call option and the price of a corresponding put option with the same strike.

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Question 26

10 marks

The shift from Over-the-Counter (OTC) trading to Central Clearing for many standardized derivatives was a major regulatory response to the 2008 financial crisis.

1. Describe three key differences between the OTC market and exchange-traded markets.
2. Discuss the advantages and disadvantages of centralized clearing from the perspective of a large commercial bank.

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